www.ijprems.com
editor@ijprems.com

INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)
(Int Peer Reviewed Journal)
Vol. 06, Issue 04, April 2026, pp : xx-xx

e-ISSN : 2583-1062
Impact Factor : 7.001

Decentralized Electronic Health Record System Using Blockchain for Secure and Transparent Healthcare

Rohit Phalke[1], Poorva Chavan[2], Abhijeet Kharade[3], Kartik Lomte[4], Prof. V. B. Deokate[5]

[1234]Department of Information Technology Engineering, SVPM's College of Engineering, Malegaon BK, Maharashtra, 413115, India.

[5]Assistant Professor, Department of Information Technology Engineering, SVPM's College of Engineering, Malegaon BK, Baramati, Maharashtra, 413115, India.

ABSTRACT

Due to the increasing use of digital technology in healthcare institutions, there have been many innovations in the storage, management, and exchange of data related to patients. One such innovation is the development of electronic health record (EHR). However, traditional EHRs are largely centralized, which makes them susceptible to various risks including breaches, unauthorized access, lack of interoperability, and poor management of personal health information by patients. This paper outlines a Decentralized Electronic Health Record System using blockchain technology.

This system uses Ethereum blockchain, Solidity smart contracts, InterPlanetary File System, and Metamask authentication protocol in managing EHRs. Patients become the authority within this system and are able to provide access or deny access to their medical records. The use of smart contracts allows for automatic user registration, access policies enforcement, and audit trails creation. Medical records will be saved in the IPFS and referenced through hashes in the blockchain network. In addition, IPFS makes sure of immutability and traceability.

Implementation of this system was done using Ganache as the local blockchain. The system provides a real-time interface between a user and his/her medical records through Metamask wallet authentication. The benefits of the system include improved data security and transparency, elimination of dependency on centralized servers, and effective management of medical data through patient-centric access control policy. Based on the experimental results, the proposed system is more trustworthy, accessible, and reliable than the current centralized EHRs.

Keywords: Blockchain, Electronic Health Record (EHR), Decentralized Healthcare, Smart Contracts, InterPlanetary File System (IPFS), Ethereum, Data Security, Patient-Centric Access Control.

1. INTRODUCTION

The healthcare industry has experienced considerable changes related to digital transformation and application of electronic solutions, including the introduction of Electronic Health Records (EHR) systems. In general, the systems allow efficient storage, handling, and exchange of patient medical data. However, EHR systems tend to be centralized. As a result, several issues are associated with such platforms, including potential data breaches, single point failures, low interoperability, and lack of patients' control over their data.

In centralized EHR systems, data of patients is stored in servers owned either by hospitals or third parties. Such a framework increases the likelihood of privacy violations since data may be misused or accessed by individuals without proper authorizations. Furthermore, patients do not control access to their medical records. Thus, there are problems related to privacy, security, and data transparency.

Blockchain technology provides an opportunity to resolve the described problems and design highly secure, patient-oriented and transparent systems. Blockchain is a type of database that is distributed and decentralized. It implies that all transactions will be recorded immutably and that data cannot be modified. Smart contracts help manage permissions and access and prevent malicious attempts. In general, patients will have full access and ownership to their medical data and will decide whether it is appropriate to share it with others.

This paper aims to describe and evaluate a Decentralized Electronic Health Record System implemented with the use of blockchain, IPFS, and smart contracts. In particular, the paper will focus on the development process, the advantages of using blockchain in EHR systems, and how the proposed system is better than traditional EHR solutions. Therefore, this project may prove effective for improving patient experiences and enhancing data security.

2. LITERATURE SURVEY

[1] A. Shahnaz, U. Qamar, and A. Khalid, "Using Blockchain for Electronic Health Records," IEEE Access, 2019. This paper examines the application of blockchain technology to increase the security of electronic health records. The authors discuss how the use of the blockchain framework helps protect medical record references and maintain their immutability and transparency. By using blockchain, the authors ensure the security of records and increase the level of trust between healthcare entities due to the impossibility of unauthorized records alteration. The key benefit of blockchain in healthcare applications includes the use of decentralization and cryptography. Still, this approach provides access control to the patient only to some extent.

Drawback: The system suggested by the authors does not allow for full real-time patient-driven permission management.

[2] M. M. Madine et al., "Blockchain for Giving Patients Control Over Their Medical Records," IEEE Access, 2020. The paper focuses on the creation of a blockchain-based system designed to provide increased patient involvement in controlling the exchange of healthcare information. By introducing smart contracts for managing access policies, the authors have provided improved access control and increased transparency. Still, the access models are predetermined and involve some processes outside the blockchain network.

Drawback: The suggested system fails to include an independent permission management function allowing patients to give or revoke access rights at any time.

[3] S. Nagpal et al., "Medical Record Management System using Blockchain Technology," IEEE Xplore, 2023. This article discusses a blockchain-based medical record management system. The authors demonstrate its ability to provide enhanced security through the creation of references and immutability. They emphasize the ability of blockchain to track the history of patients' medical data in an unalterable way. Nonetheless, there is no clear division of permissions according to roles.

Drawback: Without a strict access separation based on the roles (doctors, diagnostic centers), the patient may face the problem of too broad access permissions to data.

[4] A. Al-Mamun, S. Azam, and C. Gritti, "Blockchain-Based Electronic Health Records Management: A Comprehensive Review and Future Research Direction," IEEE Access, 2023.

The paper provides a detailed analysis of several blockchain-enabled solutions to EHR management problems and offers some directions for further development. Scalability, access control, and data protection are the main issues identified in this review. While discussing the benefits of blockchain in general, the authors note that most systems lack complete auditability and flexible permissions.

Drawback: The majority of the analyzed systems cannot be considered fully auditable in terms of user actions (permissions management, reports uploading, etc.).

[5] S. Rekik, N. Alsulaiman, and N. Albadrani, "A Health Record Management System Using Smart Contract and Blockchain," IEEE Xplore, 2024.

The paper presents the development of the blockchain-enabled system aimed at improving access control in healthcare through smart contracts. The authors note the positive influence of blockchain on overall data security and improved trust. Nonetheless, the access policies are partly implemented through the off-chain components.

Drawback: Partial permission enforcement reduces transparency and reliability.

3. PROPOSED SYSTEM

The proposed system follows a multi-layered architecture that integrates blockchain technology, decentralized storage, and web-based interfaces to provide a secure and patient-centric Electronic Health Record (EHR) system.

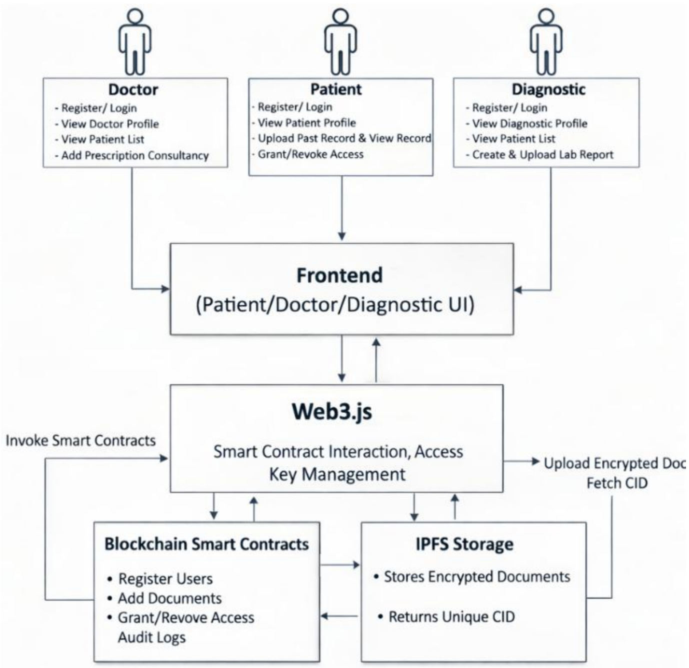


Figure 1: System Architecture

Architecture Components

A. User Layer

The system is designed around three primary user roles, each with distinct responsibilities and controlled access privileges. The patient acts as the central authority and owner of medical data, having full control over uploading records, granting or revoking access permissions, and viewing all associated medical reports. The doctor is authorized to access patient records only after receiving explicit permission and can contribute by adding consultation details such as diagnosis and prescriptions. The diagnostic center is responsible for uploading laboratory reports upon receiving patient consent, without having unrestricted access to complete patient records. Each user interacts with the system through a dedicated interface tailored to their role, ensuring usability while maintaining strict access cont

B. Frontend Layer

The frontend layer provides an integrated and user-friendly interface that enables interaction between users and the underlying blockchain infrastructure. It supports essential functionalities such as user registration, authentication, medical record upload, permission management, and data retrieval. The interface is designed to offer role-based dashboards for patients, doctors, and diagnostic centers, ensuring that each user accesses only the features relevant to their role. By abstracting the complexity of blockchain operations, the frontend ensures seamless usability while securely communicating with the decentralized backend.

C. Web3.js Layer

The Web3.js layer functions as a middleware that connects the frontend application to the blockchain network. It facilitates communication with smart contracts by enabling transaction creation, submission, and execution on the Ethereum blockchain. This layer also manages wallet-based authentication through Metamask, allowing secure user identity verification without relying on traditional password mechanisms. Additionally, it handles access key interactions and ensures that all critical system operations, such as registration, permission updates, and report submissions, are executed as verified blockchain transactions.

D. Blockchain Smart Contracts

Smart contracts deployed on the Ethereum blockchain form the core logic of the proposed system. These contracts are responsible for managing user registration, mapping identities to wallet addresses, enforcing access control policies, and maintaining a comprehensive audit trail of all system activities. They also establish a secure linkage between stored medical data and its corresponding references on the blockchain. By automating these processes, smart contracts eliminate the need for intermediaries and ensure that all operations are executed in a transparent, immutable, and tamper-proof manner.

E. IPFS Storage Layer

The InterPlanetary File System (IPFS) is utilized as a decentralized storage solution for managing medical records, including reports and prescriptions. Instead of storing large data files directly on the blockchain, the system uploads encrypted medical documents to IPFS, which generates a unique Content Identifier (CID) for each file. This CID is then stored on the blockchain as a reference to the actual data. This approach significantly improves system scalability and efficiency while preserving data integrity, security, and decentralization. By combining IPFS with blockchain technology, the system ensures reliable and distributed storage of sensitive healthcare information.

4. METHODOLOGY

A. Design Approach

The proposed system is designed using a decentralized architectural approach to eliminate reliance on centralized healthcare data management systems. The primary objective of this design is to ensure data security, integrity, transparency, and patient-centric control over medical records. Blockchain technology is utilized to maintain an immutable and distributed ledger of all transactions, thereby preventing unauthorized data modification and ensuring trust among stakeholders. To address the limitations of storing large medical files on-chain, the system integrates the InterPlanetary File System (IPFS) for off-chain storage, enabling efficient handling of medical documents. Furthermore, the system adopts a patient-driven model in which individuals act as the sole owners of their health data, with the authority to grant or revoke access permissions dynamically. Authentication is achieved through blockchain wallet addresses, eliminating traditional password-based vulnerabilities and enhancing overall system security.

B. System Model

The system model is structured around three primary entities: patients, doctors, and diagnostic centers, each assigned specific roles and responsibilities within the decentralized environment. Patients function as the data owners and have complete control over their medical records, including uploading documents and managing access permissions. Doctors act as authorized entities who can view patient records and provide consultation reports only after obtaining explicit consent. Diagnostic centers are responsible for uploading laboratory and test reports, also contingent upon patient authorization. The system implements a role-based access control mechanism enforced through smart contracts, ensuring that each participant can only perform actions permitted by their role. All interactions between these entities occur through a decentralized application interface, which communicates with the blockchain to validate permissions and execute transactions securely.

C. Smart Contract Design

Smart contracts form the core functional component of the proposed system and are developed using the Solidity programming language on the Ethereum blockchain. These contracts are responsible for automating system operations and enforcing access control policies without the need for intermediaries. The smart contract framework includes modules for user registration, where identity details and roles are recorded on the blockchain, as well as access control mechanisms that allow patients to grant or revoke permissions dynamically. Additionally, the contracts manage the storage of IPFS-generated Content Identifiers (CIDs), which serve as references to medical records stored off-chain. Every transaction, including record uploads, access approvals, and report additions, is logged within the smart contract, ensuring transparency and traceability. This design guarantees that only authorized users can access or modify data, thereby maintaining system integrity.

D. Security Mechanisms

The proposed system incorporates multiple layers of security to protect sensitive healthcare data and ensure reliable system operation. Decentralization eliminates single points of failure, reducing the risk of large-scale data breaches. The inherent immutability of blockchain technology ensures that once data is recorded, it cannot be altered or deleted, thereby preserving data integrity. User authentication is achieved through cryptographic wallet signatures via Metamask, providing a secure and password-less authentication mechanism. Access to medical records is strictly governed by smart contract–based permission control, ensuring that only authorized entities can view or modify data. Additionally, by storing medical files on IPFS and only maintaining their hash references on the blockchain, the system achieves efficient storage while maintaining data authenticity and integrity. Together, these mechanisms establish a secure, transparent, and tamper-resistant environment for managing electronic health records.

5. IMPLEMENTATION

A. User Registration and Authentication

The system implements a secure and decentralized user registration and authentication mechanism using blockchain technology. Each user such as patient, doctor, or diagnostic center is required to register by providing relevant identity details along with their unique blockchain wallet address. Patients are registered using personal information such as name, blood group, and Aadhaar number, while doctors provide professional credentials including medical license number, specialization, and affiliated hospital. Diagnostic centers register using institutional details and authorized identification numbers.

All registration data is recorded on the Ethereum blockchain through smart contract transactions, ensuring immutability and authenticity of user identities. Authentication is performed using Metamask, which enables users to log in via wallet-based verification instead of traditional username-password combinations. This approach enhances security by leveraging cryptographic signatures, eliminating the risks associated with password theft and centralized authentication systems. Upon successful authentication, users are granted access to role-specific dashboards within the decentralized application.

B. Patient Dashboard and Record Management

The patient dashboard serves as the central control interface for managing personal health records. Patients can view their profile information stored on the blockchain and upload medical documents such as prescriptions, reports, and test results. These documents are stored in the InterPlanetary File System (IPFS), which generates a unique Content Identifier (CID) for each file. The CID is then recorded on the blockchain through smart contracts, ensuring secure linkage and data integrity.

A key feature of the patient dashboard is access control management, which allows patients to grant or revoke permissions for doctors and diagnostic centers. These permissions are enforced through smart contracts, ensuring that only authorized entities can access or upload medical data. Patients can also view a complete history of their medical records along with timestamps, providing transparency and traceability. All operations performed by the patient require confirmation through Metamask, ensuring secure and verifiable transaction execution.

C. Doctor Dashboard

The doctor dashboard provides authorized medical professionals with access to patient data and consultation functionalities. Once registered and authenticated, doctors can view their professional profiles linked to their license credentials stored on the blockchain. Doctors are only able to access patient records when explicit permission has been granted by the patient.

Upon receiving access, doctors can retrieve patient medical data using the IPFS CID stored on the blockchain. The dashboard allows doctors to analyze patient history and generate consultation reports, including diagnoses and prescriptions. These reports are recorded on the blockchain as transactions, ensuring immutability and accountability. The system ensures that doctors cannot access or modify any patient data without authorization, thereby maintaining strict data privacy and security standards.

D. Diagnostic Center Dashboard

The diagnostic center dashboard is designed to facilitate secure upload and management of laboratory reports. Diagnostic centers can register and authenticate using their institutional credentials and blockchain wallet address. Similar to doctors, diagnostic centers can only interact with patient data after receiving explicit permission from the patient.

Once authorized, diagnostic centers can upload diagnostic reports such as lab test results to IPFS. The generated CID is then stored on the blockchain, creating a secure reference to the uploaded document. Unlike doctors, diagnostic centers are restricted from viewing or modifying patient records unless explicitly permitted, ensuring minimal data exposure and adherence to privacy principles.

All report uploads are validated through blockchain transactions, providing a transparent and tamper-proof record of activities. This ensures accountability and trust within the healthcare data exchange process.

6. RESULT AND DISCUSSION

The proposed decentralized Electronic Health Record (EHR) system was successfully implemented using Ethereum blockchain, Solidity smart contracts, IPFS storage, and a web-based frontend interface integrated with Metamask. The system was deployed and tested in a local blockchain environment using Ganache, demonstrating real-time interaction between users and the blockchain network.

A. User Registration and Authentication

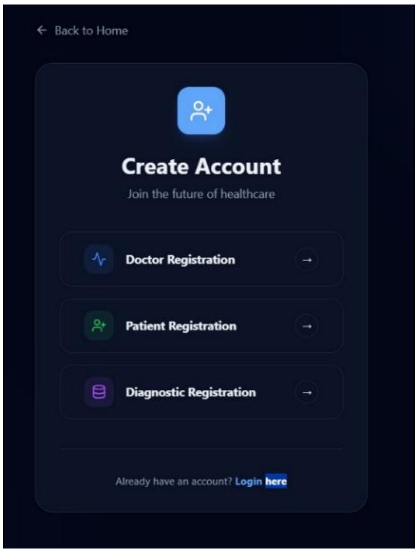


Figure 2: User Registration

The system provides a unified registration interface that allows users to register as patients, doctors, or diagnostic centers. As shown in **Fig. 2**, users can select their role and submit relevant details. Registration is completed through blockchain transactions, ensuring that user identities are securely recorded and cannot be altered. Authentication is handled through Metamask, enabling secure wallet-based login without relying on traditional password systems.

B. Patient Dashboard and Record Management

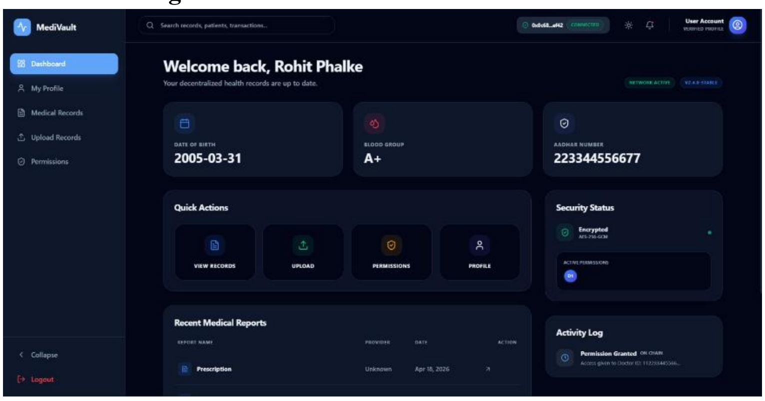


Figure 3: Patient Dashboard

The patient dashboard, illustrated in **Fig. 3**, acts as the central control panel for managing medical data. Patients can view their profile, access stored records, and monitor system activity. The dashboard also displays security status and active permissions, providing transparency and control over data sharing.

C. Medical Record Upload and Blockchain Interaction

www.ijprems.com
editor@ijprems.com

INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)
(Int Peer Reviewed Journal)
Vol. 06, Issue 04, April 2026, pp : xx-xx

e-ISSN : 2583-1062
Impact Factor : 7.001

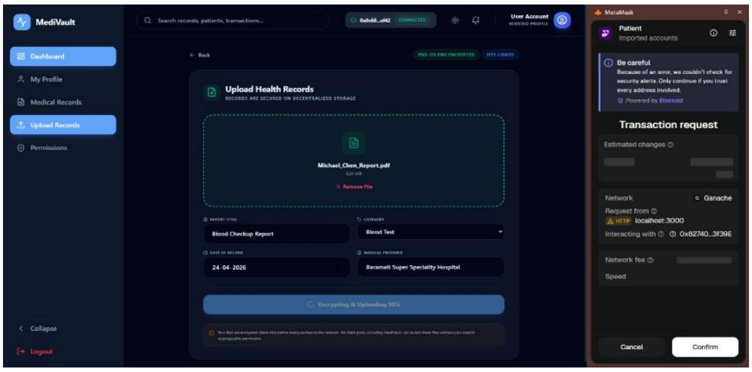


Figure 4: Medical Record Upload

As shown in **Fig. 4**, patients can upload medical records, which are first encrypted and then stored on IPFS. The system generates a unique CID that is recorded on the blockchain. Each upload requires confirmation through Metamask, ensuring that all operations are securely validated and authorized by the user.

D. Access Control Mechanism

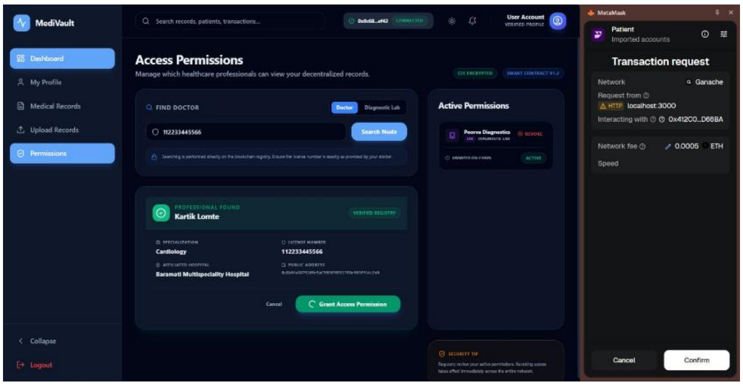
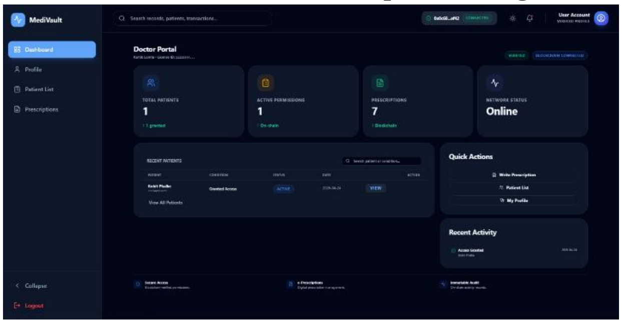


Figure 5: Access/Revoke Permission

The access control module, depicted in **Fig. 5**, allows patients to grant or revoke permissions dynamically. Permissions are enforced through smart contracts, ensuring that only authorized users can access medical data. The system also maintains a real-time list of active permissions, enhancing transparency and accountability.

E. Doctor Dashboard and Prescription Management



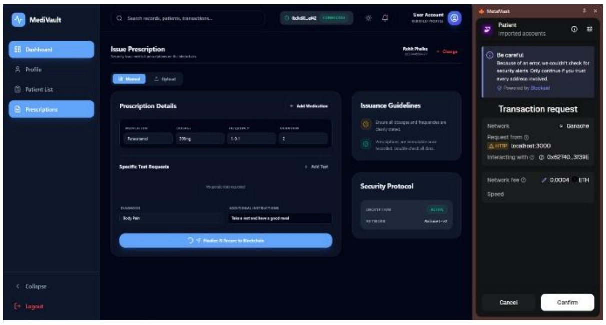
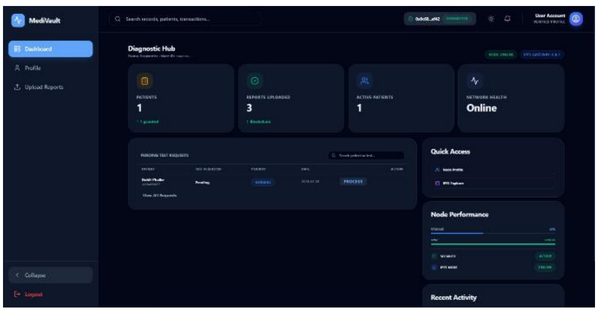


Figure 6: Doctor Dashboard **Figure 7:** Generate Prescription

The doctor dashboard, shown in **Fig. 6**, enables healthcare professionals to view authorized patient records and provide consultation services. Doctors can generate prescriptions, shown in **Fig.7**, and store them on the blockchain, ensuring that medical records remain immutable and verifiable.

F. Diagnostic Center and Report Upload



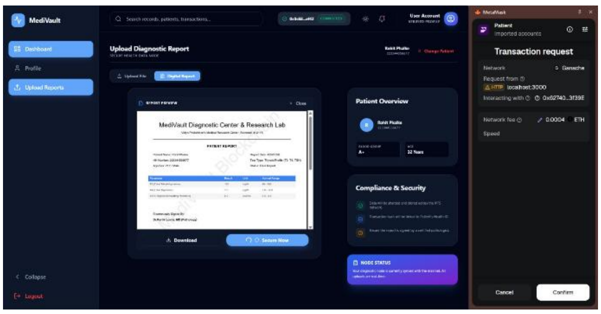


Figure 8: Diagnostic Dashboard **Figure 9:** Create & Upload Record

As illustrated in **Fig. 7 & 8**, diagnostic centers can upload lab reports after receiving patient authorization. The reports are stored on IPFS, and their references are maintained on the blockchain. This ensures secure data storage while maintaining efficient retrieval.

7. CONCLUSION

A decentralized Electronic Health Record (EHR) system, making use of blockchain, smart contracts, and IPFS, has been presented in this paper as a solution to overcome several crucial issues in current health data management systems. The suggested framework ensures that data integrity and safety are ensured using Ethereum smart contracts and IPFS for the management of access and auditing capabilities. It is also capable of implementing a patient-centric system whereby patients can exercise complete control over their health data such as allowing or denying access to medical practitioners and diagnostic centers. Metamask has been used as a security measure since it makes the process password-free.

As evidenced from the implementation and testing of the EHR system, its efficiency lies in the security, integrity, and transparency of the data being managed. The role-based access management scheme guarantees that no unauthorized person accesses critical data and ensures privacy for all parties involved in the healthcare transaction. Although the system has been developed under the restrictions of a private blockchain, without any encryption, and lacks scalability optimization, this research still lays an excellent groundwork for more advanced work in this field.

8. REFERENCES

[1] A. Shahnaz, U. Qamar, and A. Khalid, “Using blockchain for electronic health records,” *IEEE Access*, vol. 7, pp. 147782–147795, 2019.

[2] M. M. Madine, A. A. Battah, I. Yaqoob, K. Salah, R. Jayaraman, Y. Al-Hammadi, S. Pesic, and S. Ellahham, “Blockchain for giving patients control over their medical records,” *IEEE Access*, vol. 8, pp. 193102–193117, 2020.

[3] S. Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System. 2008, pp. 1–9.

[4] E. Chukwu and L. Garg, ‘‘A systematic review of blockchain in health care: Frameworks, prototypes, and implementations,’’ IEEE Access, vol. 8, pp. 21196–21214, 2020.

[5] S.Khezr, A.Yassine and R.Benlamri,‘‘Blockchain technology in health care: A comprehensive review and directions for future research,’’ Appl. Sci., vol. 9, no. 9, p. 1736, 2019.

[6] S. Nagpal, P. K. Aggarwal, K. Gupta, S. Shalini, P. Jain, and A. Agarwal, “Medical record management system using blockchain technology,” in *Proc. IEEE Xplore*, 2023.

[7] A. Al-Mamun, S. Azam, and C. Gritti, “Blockchain-based electronic health records management: A comprehensive review and future research direction,” *IEEE Access*, vol. 11, pp. 1–20, 2023.

[8] S. Rekik, N. Alsulaiman and N. Albadrani, “A health record management system using smart contract and blockchain,” in *Proc. IEEE Xplore*, 2024.

[9] A. Haddad, M. H. Habaebi, M. R. Islam, N. F. Hasbullah, and S. A. Zabidi, “Systematic review on AI-blockchain based e-healthcare records management systems,” *IEEE Access*, vol. 10, pp. 1–15, 2022.

[10] A. P. Gaigol, V. S. Wadne, S. R. Bhandari, R. S. Deshpande, and T. R. Sangole, “A systematic review on medical health data management using blockchain technology,” 2023.

[11] M. Usman and U. Qamar, “Secure electronic medical records storage and sharing using blockchain technology,” *ScienceDirect*, 2019.

[12] L. G. Careline and T. Godhavari, “Implementation of electronic health record and health insurance management system using blockchain technology,” *IJA CSA*, 2022.

[13] M. Reisman, ‘‘EHRs: The challenge of making electronic data usable and interoperable.,’’ PT, vol. 42, no. 9, pp. 572–575, Sep. 2017. [7] W. W. Koczkodaj, M. Mazurek, D. Strzałka, A. Wolny-Dominiak, and M. Woodbury-Smith, ‘‘Electronic health record breaches as social indica tors,’’ Social Indicators Res., vol. 141, no. 2, pp. 861–871, Jan. 2019.

[14] EHRs Have Made it Easy for Cardiologists to Treat Their Patients. Accessed: Jul. 8, 2020. [Online]. Available: http://tbrcinfo.blogspot.com/ 2018/12/ehrs-have-made-it-easy-for.html

[15] L. J. Kish and E. J. Topol, ‘‘Unpatients—Why patients should own their medical data,’’ Nature Biotechnol., vol. 33, no. 9, p. 921, 2015. [13] Y.B.Perez.(2015).

[16] Medical Records Project Wins Top Prize at Blockchain Hackathon. Accessed: Mar. 15, 2020. [Online]. Available: https://www. coindesk.com/medvault-wins-e5000-at-deloitte-sponsored-bloc kchain hackathon.

[17] W. J. Gordon and C. Catalini, ‘‘Blockchain technology for healthcare: Facilitating the transition to patient-driven interoperability,’’ Comput. Struct. Biotechnol. J., vol. 16, pp. 224–230, Jan. 2018.

[18] A. Boonstra, A. Versluis, and J. F. J. Vos, ‘‘Implementing electronic health records in hospitals: A systematic literature review,’’ BMC Health Services Res., vol. 14, no. 1, Sep. 2014, Art. no. 370.

[19] J. Fu, N. Wang, and Y. Cai, ‘‘Privacy-preserving in healthcare blockchain systems based on lightweight message sharing,’’ Sensors, vol. 20, no. 7, p. 1898, Mar. 2020.

[20] S.Tanwar, K.Parekh and R.Evans,‘‘Blockchain-based electronic health care record system for healthcare 4.0 applications,’’ J. Inf. Secur. Appl., vol. 50, Feb. 2020, Art. no. 102407.